Flag transitivity need not descend from a halving

Human Compression project: N014 recovery

Theorem A in the introduction of Piedade and Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1, asserts that the halved geometry is flag-transitive if and only if the original geometry is. We give a counterexample to the converse. The forward assertion in the detailed Proposition 3.14 is unaffected. This example is not thin; it is separate from the regular counterexample to residual connectivity.

Lemma 1. *Let R be a simple graph on V, and let g be an automorphism with $g^2 = 1$. Suppose x is not adjacent to gx for every $x \in V$. Define Q by $x \sim_Q y$ if and only if $x \sim_R gy$. Then Q is simple and undirected. The canonical bipartite double covers of Q and R are isomorphic by the identity on the first shore and by g on the second shore.*

Proof. Symmetry follows from $x \sim_R gy$ if and only if $gx \sim_R y$, and the hypothesis excludes a loop. In the double cover, x on the first shore is incident with y on the second precisely when $x \sim_R gy$. Thus the stated shore relabeling is the asserted type-preserving isomorphism. □

Take V to be the two-element subsets of $\{1, 2, 3, 4, 5\}$, let R have adjacency given by disjointness, and put $g = (12)$. Every $X \in V$ meets gX: if X contains exactly one of $1, 2$, its other element is fixed by g, and otherwise $X = gX$. Lemma 1 therefore applies. It is useful to write

$$a = \{1, 2\}, \quad p_i = \{1, i\}, \quad q_i = \{2, i\}, \quad b_i = \{3, 4, 5\} \setminus \{i\} \qquad (i = 3, 4, 5).$$

The complete edge list of Q is

$$\{p_i, p_j\},\ \{q_i, q_j\}\ (i < j), \qquad \{a, b_i\},\ \{b_i, p_i\},\ \{b_i, q_i\}\ (i = 3, 4, 5).$$

Indeed these are exactly the pairs X, Y for which $X \cap gY$ is empty, by separating the four displayed kinds of vertices. Thus Q is a connected cubic graph on ten vertices, with fifteen edges. The p_i and the q_i each form a triangle; a belongs to no triangle, since its three neighbors are the pairwise nonadjacent b_i. In particular Q is nonbipartite and not vertex-transitive.

Define a rank-three incidence geometry Γ as follows. Its objects of type 0 are the vertices of Q; its objects of type 1 are the edges of Q, with the usual endpoint incidence. Its objects of type 2 are two distinct elements z_+, z_-, each incident with every object of the other two types. Distinct objects of the same type are not incident.

Proposition 2. *The geometry Γ is residually connected and satisfies the source conditions B1 and B2. Its halving $\mathcal{P}(\Gamma)$ is residually connected and flag-transitive, whereas Γ is not flag-transitive. It has 60 chambers, as does its halving.*

Proof. Every flag extends to a chamber (v, e, z_ϵ) with $v \in e$. At z_ϵ the rank-two residue is the vertex-edge incidence graph of Q, hence is connected. At a vertex it is $K_{3,2}$; at an edge it is $K_{2,2}$. The whole incidence graph is connected because either type-2 object is incident with all lower objects. These are every residue of rank at least two, proving residual connectivity. Simplicity of Q is B1. Every type-2 object is incident with both endpoints of every edge and with that edge, so B2 holds. Flag transitivity would imply vertex transitivity of Q, which its triangle distinction excludes. Finally there are $15 \cdot 2 \cdot 2 = 60$ chambers.

We now identify the halving directly from its definition. Its types 0 and 1 are two copies of V, with X on the first shore incident with Y on the second when $X \cap gY = \varnothing$. The shadow of either z_ϵ is the whole connected nonbipartite graph Q. Its canonical double cover is connected, so z_ϵ has one lifted object, incident with every object of the first two types. Thus $\mathcal{P}(\Gamma)$ is the join of the bipartite double cover of Q with $\{z_+, z_-\}$.

For completeness, connectedness of that double cover follows without an enumeration. In a connected nonbipartite graph, a walk between two prescribed vertices can have either parity: traverse a path to an odd cycle, go around that cycle, and return along the same path to reverse the parity. Lifting these walks connects either choice of sheet over either vertex. The residues in the new geometry are therefore connected: at z_ϵ use this double cover, and at either lower object the residue is $K_{3,2}$. These again exhaust the rank-two residues.

After the second-shore relabeling $Y \mapsto gY$, the lower incidence relation is simply disjointness of two-element subsets. The group S_5 is transitive on ordered pairs of disjoint two-element sets: map the first pair, the second pair, and their unique remaining element separately. Consequently its action in the original labels is

$$X_0 \mapsto (\sigma X)_0, \qquad Y_1 \mapsto (g\sigma g Y)_1 \qquad (\sigma \in S_5).$$

It is transitive on the thirty incident lower pairs. Independently interchanging z_+ and z_- proves transitivity on all sixty chambers. Since every flag extends to a chamber, this is flag transitivity. The chamber count is $30 \cdot 2 = 60$. □

All nonempty higher shadows here equal Q and are nonbipartite, so the corrected parity condition is satisfied. The counterexample therefore does not exploit the separate failure of unconditional residual connectivity. Its only target is the reverse implication in the introductory flag-transitivity equivalence. The converse fails because an automorphism of the bipartite incidence geometry may act differently on its two shores, and need not descend to an automorphism of Q. The construction and proof use only elementary graph and incidence arguments; no novelty or publication-priority claim is made.